

Lab 01: Exploring a Basic Network Topology

Duration: 1 hour 30 minutes | In-Lab Assessment

Tools: Cisco Packet Tracer

Objectives

By the end of this lab, students will be able to:

- Identify network “edge” and “core” devices.
- Build a LAN topology in Packet Tracer.
- Configure IP addressing and default gateways.
- Generate ICMP (ping) traffic and capture frames.

Lab Description

Part A: Packet Tracer Basics

1. Components

- Devices Palette: End Devices → PC-PT.
- Network Devices: Switches → 2960; Routers → 2901.
- Connections: Click the lightning-bolt icon for cables (copper straight-through).

2. Save Your File: File → Save As → Lab1_Topology.pkt.

Part B: Building the Topology

1. Place Devices:

- PC1: drag PC-PT (left).
- Switch: drag Switch-2960 (center).
- Router: drag Router-2901 (right of switch).
- PC2: drag PC-PT (right).

2. Connect Devices:

- PC1 (FastEthernet0) → Switch (FastEthernet0/1) using Copper Straight-Through.
- Switch (FastEthernet0/2) → Router (GigabitEthernet0/0).
- Router (GigabitEthernet0/1) → PC2 (FastEthernet0) using Copper Straight-Through.

Part C: IP Configuration

The IP is configured on the interface.

1. PC1:

- Open PC1 → Desktop → IP Configuration:
 - IP: 192.168.1.10

- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

2. Router:

In Packet Tracer, click Router → **CLI** tab, then execute:

```

1 enable                      ! Enter privileged EXEC mode
2 configure terminal          ! Enter global configuration
   mode
3 interface GigabitEthernet0/0 ! Select interface G0/0
4   ip address 192.168.1.1 255.255.255.0 ! Assign IP
5   no shutdown               ! Bring up interface
6 exit                        ! Exit to global mode
7 interface GigabitEthernet0/1 ! Select interface G0/1
8   ip address 10.0.0.1 255.255.255.0 ! Assign IP
9   no shutdown
10 exit
11 end                        ! Return to privileged EXEC

```

3. PC2:

- Open PC2 → Desktop → IP Configuration:
 - IP: 10.0.0.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 10.0.0.1

Part E: Router-to-Router Connections (Ethernet or Serial)

Routers can be connected to each other using **Ethernet interfaces** or **Serial interfaces**. Serial interfaces are *not required* unless we are learning traditional WAN technologies.

Option 1: Router-to-Router Using Ethernet (Recommended)

- No additional modules are needed.
- Use a Copper Straight-Through cable.
- Connect GigabitEthernet0/1 of Router1 to GigabitEthernet0/1 of Router2.

Example Configuration:

```

1 ! Router1
2 enable
3 configure terminal
4   interface GigabitEthernet0/1
5     ip address 172.16.0.1 255.255.255.252
6     no shutdown
7 end
8
9 ! Router2
10 enable
11 configure terminal
12   interface GigabitEthernet0/1
13     ip address 172.16.0.2 255.255.255.252

```

```

14     no shutdown
15 end

```

This is simpler and represents modern networks. Remember, both of the routers need to be in the same network

Option 2: Router-to-Router Using Serial Ports

To add serial interfaces:

- Click Router → **Physical** tab.
- Power off router.
- Insert HWIC-2T serial module.
- Power on router.
- Connect Serial0/0/0 <-> Serial0/0/0 using DCE/DTE cable.

Example Configuration:

```

1  ! Router1
2  interface Serial0/0/0
3      ip address 172.16.0.1 255.255.255.252
4      no shutdown
5
6  ! Router2
7  interface Serial0/0/0
8      ip address 172.16.0.2 255.255.255.252
9      no shutdown

```

Part F: Adding More Ports to a Router Using Modules

Routers such as the 2901 have limited built-in interfaces. Packet Tracer allows you to add additional ports (Ethernet, Serial, etc.) using hardware modules.

Steps to Add a Module:

1. Click the router → go to the **Physical** tab.
2. Power off the router using the power switch (top-left).
3. Drag a module into an empty expansion slot:
 - HWIC-2T — adds 2 serial interfaces.
 - HWIC-4ESW — adds 4 switch-like Ethernet ports.
 - HWIC-1GE-SFP — adds 1 GigabitEthernet port.
4. Power the router back on.
5. Go to the **CLI** tab and verify new interfaces using:

```

1 show ip interface brief

```

Example: Configuring a New Ethernet Port (G2/0)

```

1 enable
2 configure terminal
3   interface GigabitEthernet2/0
4     ip address 192.168.50.1 255.255.255.0
5     no shutdown
6   exit
7 end

```

After adding modules, you can connect more PCs, routers, or switches to these interfaces.

Part G: Establishing Routing (RIP)

Configure RIP on each router via the CLI:

```

1 enable
2 configure terminal
3   router rip                                ! Enable RIP process
4   version 2                                ! Use classless RIP v2
5   network 192.168.1.0                       ! Advertise LAN on G0/0
6   network 10.0.0.0                         ! Advertise LAN on G0/1
7   network 172.16.0.0                       ! Advertise serial link subnet
8 exit
9 end

```

Configure using GUI:

- Click the router and then click on config
- Then select RIP from the left side section
- There you can see an field to add network address and a button add
- Add all the addresses a router is connected to (including the router to router network).
- Do this in both the routers

Part H: Simulation & Packet Capture

- Switch Packet Tracer to **Simulation Mode**.
- On PC1 (Desktop → Command Prompt), run:
ping 10.0.0.10 ! Send ICMP echo to PC2
- Observe the ICMP frames traversing the network and inspect each OSI layer.

Part I: Saving Configurations

To save your changes:

```

1 enable                                ! Ensure privileged EXEC mode
2 copy running-config startup-config ! Save running config to NVRAM

```

Student ID:

Lab 01 Task

In Islamic University of Technology (IUT), each computer lab has its own network. There are five labs, each with two PCs. All the networks should have the prefix 192.id, where id is the last two digits of your student id

Your task is to design an interconnection such that any PC in Lab x can reach any PC in Lab y .

Bonus: Deploy a generic server accessible only by a single lab's PCs at `http://192.168.99.10/`.

Lab 01 Grading Criteria

1. Building the topology: 15 marks

- Lab's topology - $2 * 5 = 10$ marks
- Connection between the labs = 5 marks

2. Configuring the PCs: 5 marks

- 10 PCs $\times$.5 mark each

3. Configuring the Routers: 10 marks

4. Verifying connectivity (ping): 10 marks

- Each lab's PCs local connectivity test - .5 marks each PC - 5 marks
- Global connectivity test - .5 marks each PC - 5 marks

5. Total: 40 marks

6. Bonus: 10 marks